

Fallou Diouf

Paris, France | fallou.fadal.diouf4@gmail.com | +33 7 53 24 19 39 | [GitHub](#) | [Portfolio](#) | [LinkedIn](#)

PROFESSIONAL OBJECTIVE

I am seeking a **4 to 6-month final-year internship** in Computer Vision, Deep Learning, or Machine Learning, with a particular interest in developing robust solutions for real-world applications.

EDUCATION

2025 – 2027 | Université Paris Cité, Paris, France

Master's Degree in Computer Science – Vision and Machine Intelligence

Key modules: Computer Vision, Image Analysis, 3D Imaging, Deep Learning, Pattern Recognition, Data Engineering, Signal and Image Processing.

2023 – 2024 | Université Gaston Berger, Saint-Louis, Senegal

Master 1 – Data Management and Software Engineering

2020 – 2023 | Université Gaston Berger, Saint-Louis, Senegal

Bachelor's Degree in Computer Science – Information Systems

PROFESSIONAL EXPERIENCE

Programming Tutor – C/C++, Algorithms, OOP and SQL

2023 – 2024

- Explained complex concepts from lectures, tutorials and practical sessions, adapting explanations to individual learning difficulties.
- Monitored learners' progress and supported them in developing greater autonomy.

PROJECTS

2025 – 2026 | Person Tracking with YOLOv8 and DeepSORT

Individual

Development of a system for detecting and tracking people in videos using YOLOv8 and DeepSORT. Assignment of unique identifiers (*Track ID*) and analysis of people trajectories over time.

Skills: Computer Vision, Object Detection, Object Tracking, YOLOv8, DeepSORT, OpenCV.

2025 – 2026 | Coin Detection and Classification

Team & Individual

Classical Image Processing (team): Design of a pipeline using Gaussian filtering, Canny edge detection and the Hough Transform to segment coins, combined with texture and diameter analysis methods to automate coin classification and total amount calculation.

Deep Learning (individual): Development of an AlexNet-based classifier to identify the origin and value of coins, with optimization of feature extraction.

Skills: Computer Vision, Classification, OpenCV, PyTorch, Transfer Learning, Comparative Evaluation.

2025 – 2026 | Self-Supervised Test-Time Training

Pair Project

Implementation of a *Self-Supervised Learning* approach combining SimCLR and Vision Transformer to improve the robustness of an image classification model during inference.

Validation accuracy of **90.08%** and an improvement of **+0.39 percentage points** on CIFAR-10-C with Test-Time Training.

Skills: PyTorch, SimCLR, Vision Transformer, Self-Supervised Learning, Test-Time Training.

2025 – 2026 | Information Retrieval Engine

Team Project

Design of a hybrid search engine combining lexical retrieval (BM25) and semantic search (Embeddings) for multi-domain queries.

Development of a *Reranking* module and integration of LangChain/LangGraph to improve Top-K result relevance. Best result with embeddings: **70.76% Recall@20**, **52.02% MRR** and **30.6 ms** latency.

Skills: Information Retrieval, RAG, LLM, Embeddings, LangChain, Performance Evaluation.

TECHNICAL SKILLS

Computer Vision & Image Processing

Object Detection, Object Tracking, Image Classification, Image Segmentation, Feature Extraction, Shape Recognition, Video Analysis, OpenCV, YOLOv8, DeepSORT.

Deep Learning & Machine Learning

PyTorch, TensorFlow, CNNs, AlexNet, ResNet, DenseNet, Vision Transformers, Transfer Learning, Self-Supervised Learning, SimCLR, Test-Time Training, Scikit-learn.

Information Retrieval & NLP

TF-IDF, BM25, Embeddings, BGE, RAG, LLMs, LangChain, LangGraph, Reranking, Semantic Search.

Programming, Data & Tools

Python, C, C++, JavaScript, SQL, NumPy, Pandas, PostgreSQL, Git, GitHub, Docker, Linux, FastAPI, Django, PySpark.

LANGUAGES

French: Fluent

English: Intermediate

INTERESTS

Artificial Intelligence • Scientific Research • Technology Trends • Programming